NETPILOT PROJECT GUIDE

Office / Small-Business Network

Routers, switches, VLANs per department, DHCP, NAT and inter-VLAN routing

Project owner	**Michel**
Network name	Michel's Office Network
Prepared by	NetPilot
Date	October 8, 2026
Goal	Every computer gets an IP address automatically, can message every other computer directly, and can reach the Internet.

What you will build

- One router (R1) that connects the office to the Internet and routes between departments.
- One switch (SW1) with four department VLANs plus a management VLAN.
- A DHCP server on R1 so each computer receives its address, mask, gateway and DNS automatically.
- NAT so all private office addresses share one public address on the Internet.
- A test plan that proves every computer can send messages to every other computer.

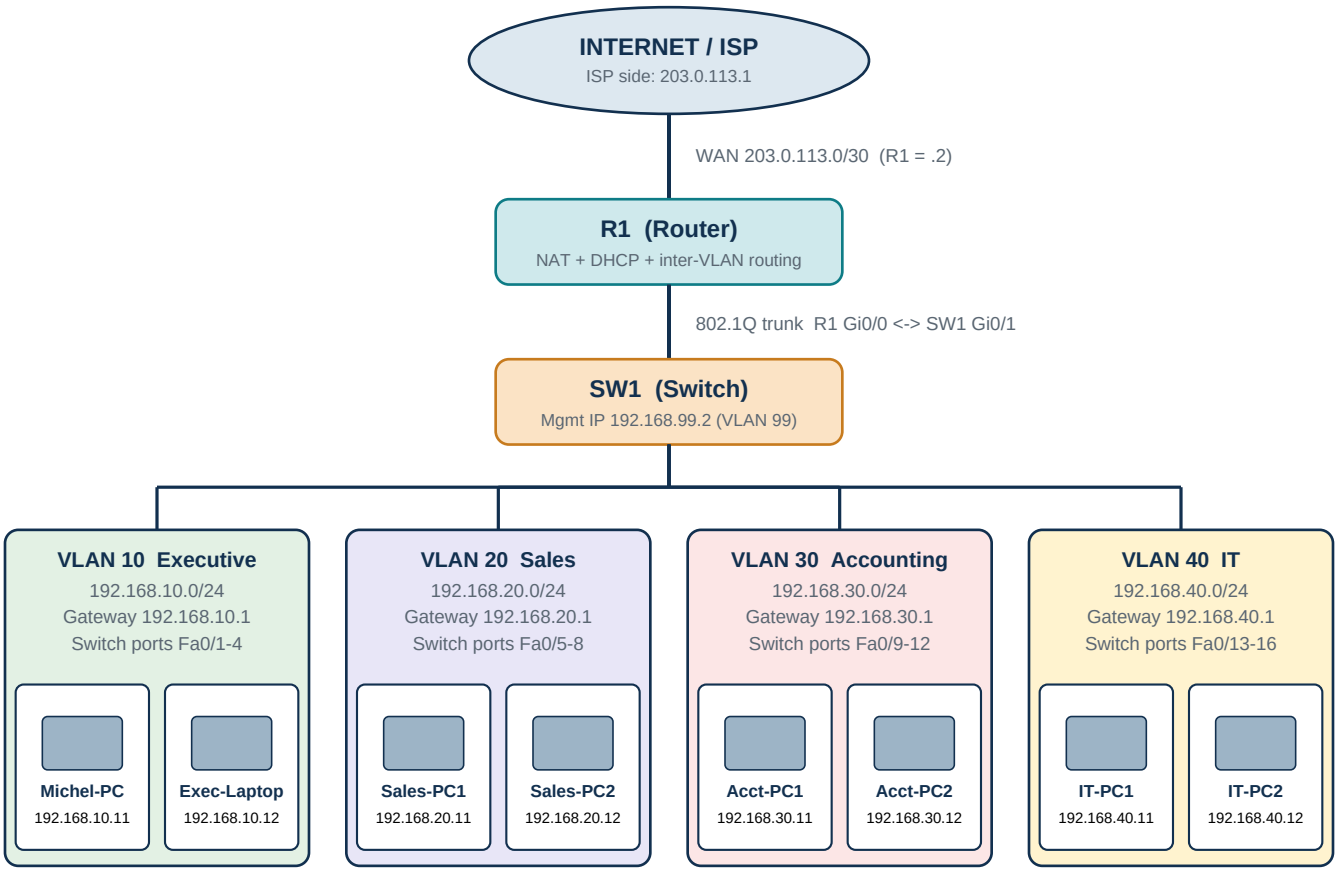
1. Why we separate the network

A flat network puts everyone in one big room: every broadcast reaches every device, and a problem on one PC can touch all the others. We split Michel's office into departments, each in its own VLAN (virtual LAN) and its own IP subnet. Departments still talk to each other, but only through the router, which is where we can later add rules.

Segment	Why it is separate
VLAN 10 Executive	Michel's and management's files and email stay in their own protected group.
VLAN 20 Sales	Handles outside customers and lots of email, so it is the most exposed to phishing. Keeping it apart limits the damage.
VLAN 30 Accounting	Payroll and financial records. A separate segment makes it easy to restrict who can reach it.
VLAN 40 IT	Administrators need tools and access that normal staff should not have.
VLAN 99 Network management	Switch and router admin addresses live here, away from user PCs.

Extra benefits: smaller broadcast domains (faster network), easier troubleshooting (a problem is limited to one VLAN), and room to grow (add a switch without redesigning).

2. Network topology



How a message travels. Inside a VLAN, the switch delivers the message directly from one port to the other using MAC addresses. Between VLANs, the message goes up the trunk to R1, R1 routes it to the destination

subnet, and it comes back down the same trunk to the right port. This design is called *router-on-a-stick*.

3. IP addressing plan

Every department gets one /24 subnet (255.255.255.0). The router is always **.1**. Addresses **.1 to .50** are reserved for infrastructure and fixed computers; DHCP hands out **.51 to .254**.

VLAN	Name	Network	Gateway (R1)	Static range	DHCP range
10	Executive	192.168.10.0/24	192.168.10.1	.2 - .50	.51 - .254
20	Sales	192.168.20.0/24	192.168.20.1	.2 - .50	.51 - .254
30	Accounting	192.168.30.0/24	192.168.30.1	.2 - .50	.51 - .254
40	IT	192.168.40.0/24	192.168.40.1	.2 - .50	.51 - .254
99	Net management	192.168.99.0/24	192.168.99.1	.2 - .50	none (static only)

Each computer and device

Device	VLAN	IP address	Mask	Gateway	Connects to
Michel-PC (owner)	10	192.168.10.11	255.255.255.0	192.168.10.1	SW1 Fa0/1
Exec-Laptop	10	192.168.10.12	255.255.255.0	192.168.10.1	SW1 Fa0/2
Sales-PC1	20	192.168.20.11	255.255.255.0	192.168.20.1	SW1 Fa0/5
Sales-PC2	20	192.168.20.12	255.255.255.0	192.168.20.1	SW1 Fa0/6
Acct-PC1	30	192.168.30.11	255.255.255.0	192.168.30.1	SW1 Fa0/9
Acct-PC2	30	192.168.30.12	255.255.255.0	192.168.30.1	SW1 Fa0/10
IT-PC1	40	192.168.40.11	255.255.255.0	192.168.40.1	SW1 Fa0/13
IT-PC2	40	192.168.40.12	255.255.255.0	192.168.40.1	SW1 Fa0/14
SW1 (management)	99	192.168.99.2	255.255.255.0	192.168.99.1	R1 via Gi0/1 trunk
R1 LAN (sub-interfaces)	10-99	.1 in every subnet	255.255.255.0	-	SW1 Gi0/1 trunk
R1 WAN (Gi0/1)	-	203.0.113.2	255.255.255.252	203.0.113.1	ISP router Gi0/0
ISP router (test)	-	203.0.113.1	255.255.255.252	-	R1 Gi0/1

The WAN range 203.0.113.0/24 is a documentation range reserved for examples. In a real office, replace it with the address your Internet provider gives you.

STEP 1 Build the topology and cable it

Open Cisco Packet Tracer (or use real equipment) and place the devices below. Use straight-through copper cables for PCs, and a straight-through cable for each router/switch link.

From	To	Cable / purpose
ISP router Gi0/0	R1 Gi0/1	WAN link to the Internet
R1 Gi0/0	SW1 Gi0/1	Trunk carrying all VLANs
SW1 Fa0/1, Fa0/2	Michel-PC, Exec-Laptop	Access ports, VLAN 10
SW1 Fa0/5, Fa0/6	Sales-PC1, Sales-PC2	Access ports, VLAN 20
SW1 Fa0/9, Fa0/10	Acct-PC1, Acct-PC2	Access ports, VLAN 30
SW1 Fa0/13, Fa0/14	IT-PC1, IT-PC2	Access ports, VLAN 40

Suggested models in Packet Tracer: Cisco 2911 (or 1941) for R1 and the ISP router, Cisco 2960-24TT for SW1, and generic PCs. If your router has no Gi0/1, use any two free ports and adjust the names in the commands.

STEP 2 Configure the switch: VLANs, access ports, trunk

Connect to SW1 and paste the commands. This creates the VLANs, places each computer's port in its department, and builds the trunk to the router.

```
enable
configure terminal
hostname SW1
banner motd # Property of Michel - authorized use only #
!
vlan 10
 name Executive
vlan 20
 name Sales
vlan 30
 name Accounting
vlan 40
 name IT
vlan 99
 name NetMgmt
exit
!
interface range fa0/1 - 4
 switchport mode access
 switchport access vlan 10
interface range fa0/5 - 8
 switchport mode access
 switchport access vlan 20
interface range fa0/9 - 12
 switchport mode access
 switchport access vlan 30
interface range fa0/13 - 16
 switchport mode access
 switchport access vlan 40
!
interface gi0/1
 description Trunk to R1
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30,40,99
```

```
!
interface vlan 99
 ip address 192.168.99.2 255.255.255.0
 no shutdown
exit
ip default-gateway 192.168.99.1
!
end
copy running-config startup-config
```

STEP 3 Configure the router: inter-VLAN routing

R1 gets one **sub-interface** per VLAN. Each sub-interface is the default gateway for its department, which is what lets VLANs talk to each other. The physical port must be turned on and must have **no** IP address of its own.

```
enable
configure terminal
hostname R1
banner motd # Property of Michel - authorized use only #
no ip domain-lookup
!
interface gi0/0
 description Trunk to SW1
 no ip address
 no shutdown
!
interface gi0/0.10
 description Executive
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
 ip nat inside
interface gi0/0.20
 description Sales
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0
 ip nat inside
interface gi0/0.30
 description Accounting
 encapsulation dot1Q 30
 ip address 192.168.30.1 255.255.255.0
 ip nat inside
interface gi0/0.40
 description IT
 encapsulation dot1Q 40
 ip address 192.168.40.1 255.255.255.0
 ip nat inside
interface gi0/0.99
 description NetMgmt
 encapsulation dot1Q 99
 ip address 192.168.99.1 255.255.255.0
 ip nat inside
!
end
copy running-config startup-config
```

STEP 4 Turn on DHCP so computers get addresses automatically

We reserve .1 to .50 in every subnet (router, switch and fixed computers) and create one pool per department. VLAN 99 is static only, so it has no pool.

```
configure terminal
ip dhcp excluded-address 192.168.10.1 192.168.10.50
ip dhcp excluded-address 192.168.20.1 192.168.20.50
ip dhcp excluded-address 192.168.30.1 192.168.30.50
ip dhcp excluded-address 192.168.40.1 192.168.40.50
!
ip dhcp pool EXECUTIVE
 network 192.168.10.0 255.255.255.0
 default-router 192.168.10.1
 dns-server 8.8.8.8
ip dhcp pool SALES
 network 192.168.20.0 255.255.255.0
 default-router 192.168.20.1
 dns-server 8.8.8.8
ip dhcp pool ACCOUNTING
 network 192.168.30.0 255.255.255.0
 default-router 192.168.30.1
 dns-server 8.8.8.8
ip dhcp pool IT
 network 192.168.40.0 255.255.255.0
 default-router 192.168.40.1
 dns-server 8.8.8.8
end
copy running-config startup-config
```

STEP 5 Configure NAT and the Internet connection

NAT (overload, also called PAT) lets every private 192.168.x.x address share R1's single public address 203.0.113.2 when going to the Internet.

```
configure terminal
interface gi0/1
 description WAN to ISP
 ip address 203.0.113.2 255.255.255.252
 ip nat outside
 no shutdown
!
access-list 1 permit 192.168.0.0 0.0.255.255
ip nat inside source list 1 interface gi0/1 overload
ip route 0.0.0.0 0.0.0.0 203.0.113.1
end
copy running-config startup-config
```

Test ISP router (only needed in a lab, so there is something to ping outside the office):

```
enable
configure terminal
hostname ISP
interface gi0/0
 ip address 203.0.113.1 255.255.255.252
 no shutdown
interface loopback0
 ip address 8.8.8.8 255.255.255.255
end
```

STEP 6 Set up each computer

Option A - fixed addresses (matches the table in section 3, recommended for the project). On each PC open *Desktop* > *IP Configuration* > *Static* and enter:

Computer	IP address	Subnet mask	Default gateway	DNS
Michel-PC	192.168.10.11	255.255.255.0	192.168.10.1	8.8.8.8
Exec-Laptop	192.168.10.12	255.255.255.0	192.168.10.1	8.8.8.8
Sales-PC1	192.168.20.11	255.255.255.0	192.168.20.1	8.8.8.8
Sales-PC2	192.168.20.12	255.255.255.0	192.168.20.1	8.8.8.8
Acct-PC1	192.168.30.11	255.255.255.0	192.168.30.1	8.8.8.8
Acct-PC2	192.168.30.12	255.255.255.0	192.168.30.1	8.8.8.8
IT-PC1	192.168.40.11	255.255.255.0	192.168.40.1	8.8.8.8
IT-PC2	192.168.40.12	255.255.255.0	192.168.40.1	8.8.8.8

Option B - automatic (DHCP). Choose *DHCP* on the PC. It receives the next free address from .51 upward, along with the mask, gateway and DNS. Use this for new staff, laptops and phones.

Both options work together because the fixed addresses (.11, .12) sit inside the excluded range, so DHCP never hands them to anyone else.

STEP 7 Test: send messages between every computer

A "message" between computers is a network packet. The ping command sends one and waits for the reply, which proves the two computers can talk. Open *Desktop* > *Command Prompt* on **Michel-PC** and run:

```
C:\> ping 192.168.10.1        (own gateway - R1)
C:\> ping 192.168.10.12       (same VLAN - direct through the switch)
C:\> ping 192.168.20.11       (Sales - routed through R1)
C:\> ping 192.168.30.11       (Accounting - routed through R1)
C:\> ping 192.168.40.11       (IT - routed through R1)
C:\> ping 8.8.8.8             (Internet - routed + NAT)
```

In Packet Tracer you can also click the *envelope* tool (*Add Simple PDU*), click the sending PC, then the receiving PC, and watch the message travel in Simulation mode. Repeat from at least one PC in every department.

> The first ping often shows one timeout while the computers learn each other's MAC addresses (ARP). That is normal. The next ones should reply.

From	To	Path	Expected
Michel-PC	Exec-Laptop	Same VLAN, switch only	Reply
Michel-PC	Sales-PC1 / Acct-PC1 / IT-PC1	Different VLAN, via R1	Reply
Sales-PC1	Acct-PC2	Different VLAN, via R1	Reply
IT-PC1	Every other PC	Via R1	Reply
Any PC	SW1 192.168.99.2	Via R1 to VLAN 99	Reply
Any PC	8.8.8.8	R1 + NAT to ISP	Reply

Confirm on the devices

```
SW1# show vlan brief                (ports sit in the right VLANs)
SW1# show interfaces trunk          (Gi0/1 trunking, VLANs 10,20,30,40,99)
R1#  show ip interface brief        (all sub-interfaces up/up)
R1#  show ip route                  (5 connected LANs + default route)
R1#  show ip dhcp binding           (leases handed out)
R1#  show ip nat translations       (appears while pinging 8.8.8.8)
```

8. Troubleshooting checklist

Symptom	Likely cause	Fix
PC gets 169.254.x.x	Port in wrong VLAN, or DHCP pool missing	Check `show vlan brief`; check the pool and excluded range on R1
PC cannot ping its gateway	Trunk down or VLAN not allowed	Check `show interfaces trunk`; confirm R1 Gi0/0 is no shutdown
Same VLAN works, other VLAN fails	Missing sub-interface or wrong encapsulation number	Match `encapsulation dot1Q` number to the VLAN ID
Office works, Internet fails	Missing default route, or NAT inside/outside swapped	Check `ip route` and the ip nat inside/outside lines
Wrong gateway on PC	Typo in static settings	Compare with the table in section 3

9. Optional hardening (after everything works)

Right now every department can reach every other, as requested. To tighten security later, add rules on R1. Example: stop Sales from reaching Accounting and the management network, but allow everything else.

```
configure terminal
ip access-list extended SALES-IN
 deny ip 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255
 deny ip 192.168.20.0 0.0.0.255 192.168.99.0 0.0.0.255
 permit ip any any
interface gi0/0.20
 ip access-group SALES-IN in
end
copy running-config startup-config
```

Other good next steps: set an `enable secret` password, enable SSH on SW1 and R1, shut down unused switch ports, and add a guest Wi-Fi VLAN.

Project sign-off

Project owner	**Michel**	Signature	
Date completed		Reviewed by	NetPilot